lab02

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1

As estatísticas nescessárias são as seguintes:

```
#n = número total de voos
#atrasados = número de voos com ARRIVAL_DELAY > 10
```

Com isso, conseguimos a porcentagem total

2

```r
library(dplyr)
```

```
Warning: pacote 'dplyr' foi compilado no R versão 4.5.3


Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

    filter, lag

Os seguintes objetos são mascarados por 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(readr)
```

```
Warning: pacote 'readr' foi compilado no R versão 4.5.3
```

```r
library(ggplot2)
```

```
Warning: pacote 'ggplot2' foi compilado no R versão 4.5.3
```

```r
getStats <- function(input, pos) {

  input %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) %>%
    filter(!is.na(DAY),
           !is.na(MONTH),
           !is.na(AIRLINE),
           !is.na(ARRIVAL_DELAY)) %>%
    group_by(DAY, MONTH, AIRLINE) %>%
    summarise(
      n = n(),
      atrasados = sum(ARRIVAL_DELAY > 10),
      .groups = "drop"
    )
}
```

3

```r
callback <- DataFrameCallback$new(getStats)

read_csv_chunked(
  "flights.csv",
  callback = callback,
  chunk_size = 100000,
  col_types = cols_only(
    DAY = col_integer(),
    MONTH = col_integer(),
    AIRLINE = col_character(),
    ARRIVAL_DELAY = col_double()
  ),
  progress = FALSE
)
```

```
# A tibble: 1,478 x 5
     DAY MONTH AIRLINE     n atrasados
   <int> <int> <chr>   <int>     <int>
 1     1     1 AA       1379       514
 2     1     1 DL       1558       132
 3     1     1 UA       1269       367
 4     1     1 US       1028       201
 5     2     1 AA       1508       649
 6     2     1 DL       2261       406
 7     2     1 UA       1411       493
 8     2     1 US       1174       229
 9     3     1 AA       1425       767
10     3     1 DL       2021       702
# i 1,468 more rows
```

4

```r
computeStats <- function(stats) {

  stats %>%
    group_by(AIRLINE, MONTH, DAY) %>%
    summarise(
      Total = sum(Total),
      Atrasados = sum(Atrasados),
      .groups = "drop"
    ) %>%
    mutate(
      Cia = AIRLINE,
      Data = as.Date(
        paste(2015, MONTH, DAY, sep = "-")
      ),
      Perc = Atrasados / Total
    ) %>%
    select(Cia, Data, Perc)

}
```

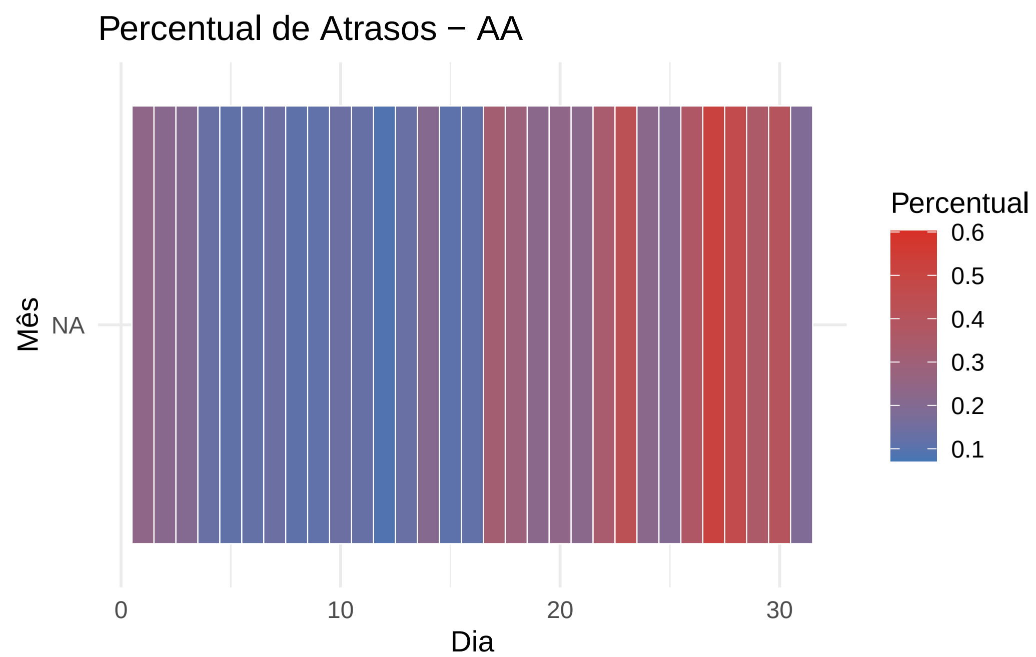
5

```r
library(readr)
library(dplyr)
library(ggplot2)

dados <- read_csv("flights.csv")
```

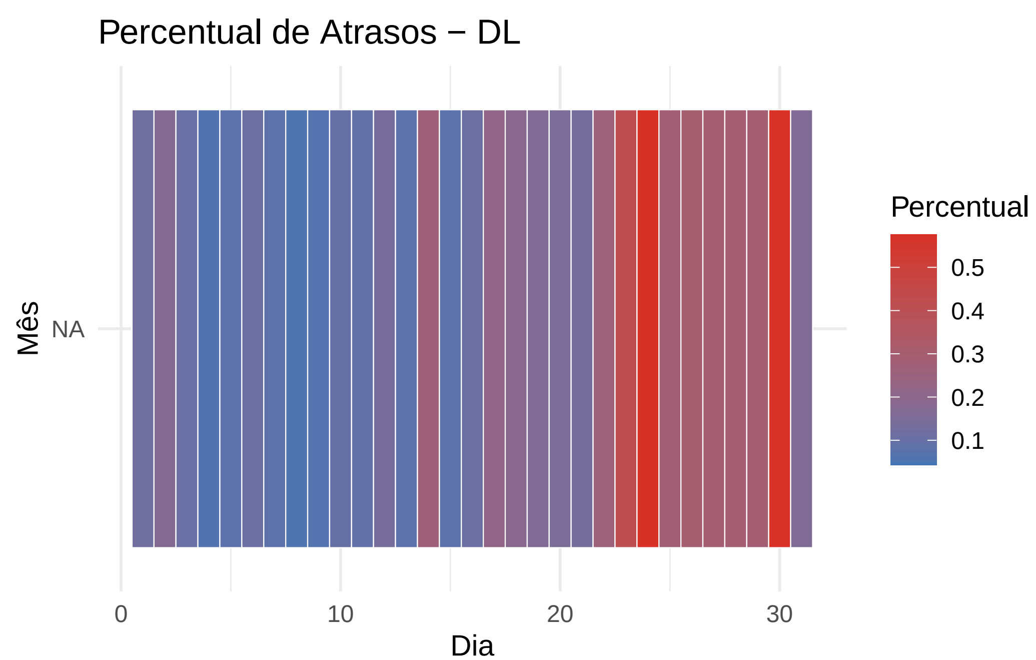
```
Rows: 5819079 Columns: 31
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (11): AIRLINE, TAIL_NUMBER, ORIGIN_AIRPORT, DESTINATION_AIRPORT, SCHEDUL...
dbl (20): YEAR, MONTH, DAY, DAY_OF_WEEK, FLIGHT_NUMBER, DEPARTURE_DELAY, TAX...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
stats <- dados %>%
  filter(AIRLINE %in% c("AA","DL","UA","US")) %>%
  filter(!is.na(ARRIVAL_DELAY)) %>%
  group_by(AIRLINE, MONTH, DAY) %>%
  summarise(
    Total = n(),
    Atrasados = sum(ARRIVAL_DELAY > 10),
    .groups = "drop"
  )

resultado <- stats %>%
  mutate(
    Cia = AIRLINE,
    Data = as.Date(paste(2015, MONTH, DAY, sep = "-")),
    Perc = Atrasados / Total
  ) %>%
  select(Cia, Data, Perc)

baseCalendario <- function(stats, cia){

  stats %>%
    filter(Cia == cia) %>%
    mutate(
      Mes = factor(format(Data,"%b"), levels = month.abb),
```

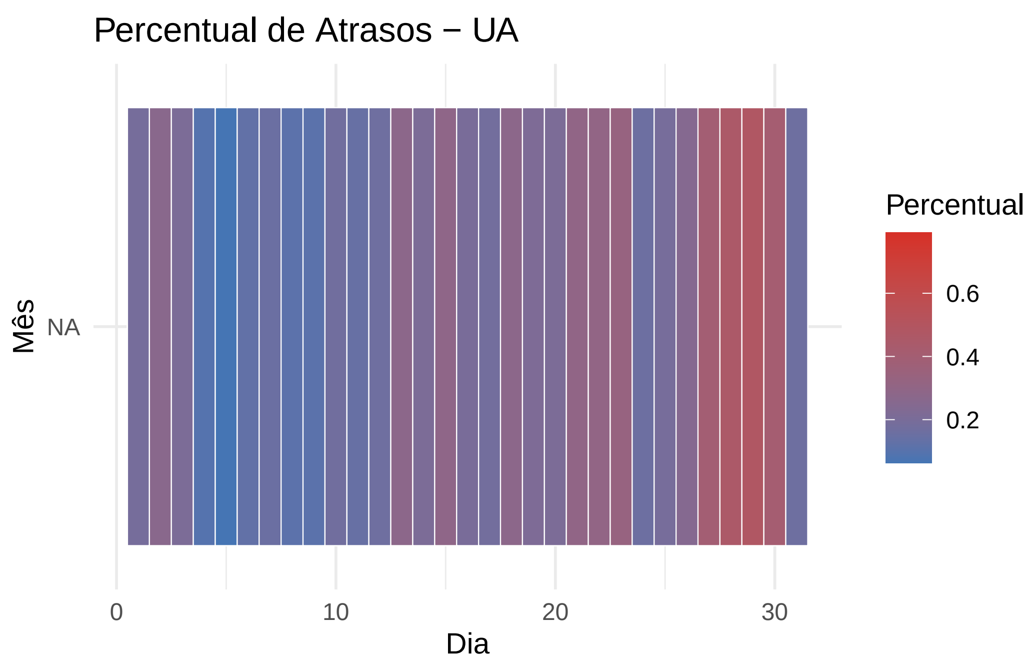
```r
      Dia = as.numeric(format(Data,"%d"))
    ) %>%
    ggplot(aes(Dia, Mes, fill = Perc)) +
    geom_tile(color = "white") +
    scale_fill_gradient(
      low = "#4575b4",
      high = "#d73027"
    ) +
    labs(
      x = "Dia",
      y = "Mês",
      fill = "Percentual"
    ) +
    theme_minimal()

}

cAA <- baseCalendario(resultado,"AA") +
  ggtitle("Percentual de Atrasos - AA")

cDL <- baseCalendario(resultado,"DL") +
  ggtitle("Percentual de Atrasos - DL")

cUA <- baseCalendario(resultado,"UA") +
  ggtitle("Percentual de Atrasos - UA")

cUS <- baseCalendario(resultado,"US") +
  ggtitle("Percentual de Atrasos - US")

cAA
```


Percentual de Atrasos − AA
Mês
NA
0
10
20
30
Dia
Percentual
0.6
0.5
0.4
0.3
0.2
0.1

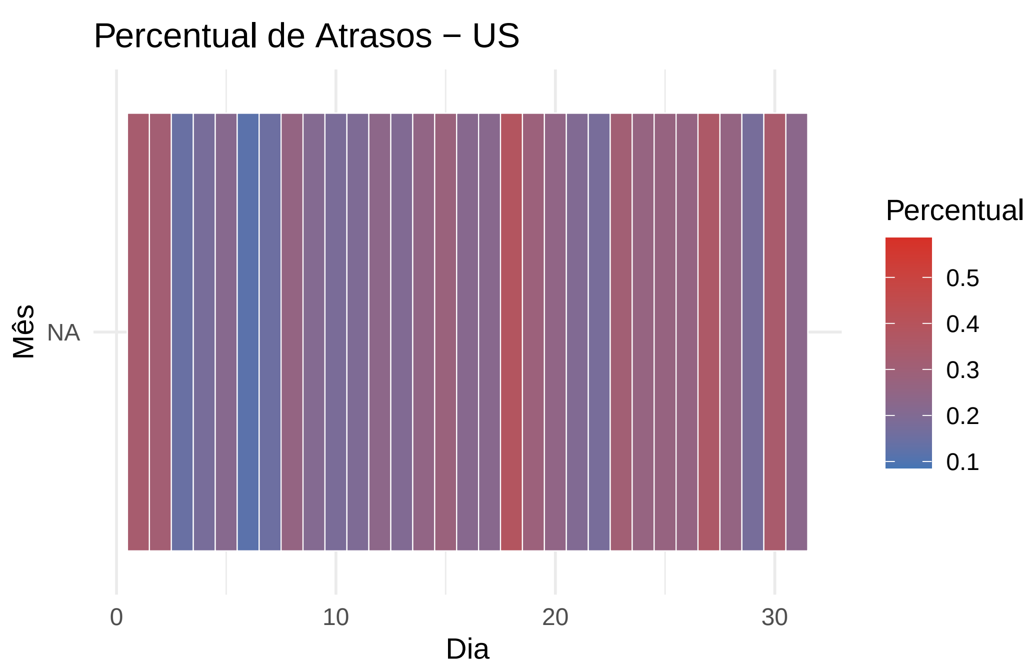
```
cDL
```


Percentual de Atrasos − DL
Mês
NA
0
10
20
30
Dia
Percentual
0.5
0.4
0.3
0.2
0.1

```
cUA
```


Percentual de Atrasos − UA
Mês
NA
0
10
20
30
Dia
Percentual
0.6
0.4
0.2

```
cUS
```


Percentual de Atrasos − US
Mês
NA
0
10
20
30
Dia
Percentual
0.5
0.4
0.3
0.2
0.1

```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-24 21:50:58 -03"
```